16.07: Dynamics

Fall 2026

Course Description

This course is about modeling and simulating mechanical systems, starting with a review of basic Newtonian mechanics of point-mass particles, then covering rigid-body and multi-body systems. Applications to aircraft, spacecraft, and robotic systems will be emphasized.

Prerequisites: Officially – (16.001 or 16.002) and (16.003 or 16.004). Practically – basic mechanics at the level of 8.01, calculus and vectors/matrices at the level of 18.02, and prior experience with a high-level programming language like Python or MATLAB (we'll use MATLAB).

Instructors

Lecturer: Zac Manchester	**Email:** zacm@mit.edu	**Office:** 31-243
TA: John Zhang	**Email:** jzhang3@mit.edu	

Logistics

- Lectures will be held **Tuesdays and Thursdays 11:00–12:30 in 35-225**. Lectures will also be recorded for later viewing.
- Recitations will be held **Fridays 3:00–4:00 in 35-225**. These will primarily be used for office hours, but three of these times will be used for labs.
- Office hours will be **Tuesdays 4:00–5:00 in 33-116** (location may change in the future).
- Homework assignments will be due **Mondays by 11:59 PM** most weeks.
- There will be a short (5 minute) quiz in class corresponding to each homework assignment. Quizzes will be in class on the Tuesday after the correspoding homeowork deadline.
- There will be an in-class mid-term exam and a cumulative final exam.
- Piazza will be used for general discussion and Q&A outside of class and office hours.

Learning Objectives

By the end of this course, students should be able to do the following:

1. Select appropriate coordinates to describe the motion of particle and multi-body system
2. Formulate dynamics in non-inertial reference frames
3. Identify and exploit conserved quantities like energy and momentum
4. Derive equations of motion for multi-body systems using the Euler-Lagrange equation
5. Model the orbital dynamics of a spacecraft as central-force motion of a particle
6. Use Euler's equation to model and analyze the attitude dynamics of a rigid body and apply this to aircraft and spacecraft

Learning Resources

There is no required textbook for this course. Video recordings of lectures and lecture notes will be posted online. The following books are good references for the topics discussed in the course that you may want to refer to:

1. J. L. Meriam, L. G. Kraige, and J. N. Bolton, Engineering Mechanics: Dynamics, 9th ed. Hoboken, NJ, USA: Wiley, 2020.
2. L. D. Landau and E. M. Lifshitz, Mechanics, 3rd ed. Oxford, U.K.: Butterworth-Heinemann, 1976.
3. H. Goldstein, C. Poole, and J. Safko, Classical Mechanics, 3rd ed. San Francisco, CA, USA: Addison Wesley, 2002.
4. K. M. Lynch and F. C. Park, Modern Robotics: Mechanics, Planning, and Control. Cambridge, U.K.: Cambridge University Press, 2017.

Assignments

- Students will be asked to complete homework assignments roughly weekly that include both an analytical question and a numerical simulation question to be completed in MATLAB.
- There will be three labs throughout the semester during the Friday recitation time. On lab weeks, homework assignments will include analysis of experiment data.

Quizzes and Exams

- Short quizzes will be given approximately weekly in class. These are intended to take no more than 5 minutes to complete.
- There will be an in-class mid-term exam and a cumulative final exam.

Grading

Grading will be based on:

- 25% Homework and Labs
- 15% Quizes
- 25% Mid-Term Exam
- 30% Final Exam
- 5% Participation

Students can earn a full participation grade through any combination of office hours, Piazza discussions, recitation attendance, and by offering constructive feedback about the course to the instructors.

Course Policies

Late Assignments: Students are allowed a budget of six late days for turning in homework with no penalty throughout the semester. They may be used together on one assignment, or separately across multiple assignments. Beyond these six days, no other late homework will be accepted.
Quiz Grades: The lowest quiz grade will be dropped from each student's semester cumulative grade. This can also be used to cover an absence.
Collaboration: Students are encouraged to collaborate on labs and homework assignments, but each student must turn in their own assignment by the deadline.
AI Use: Students are permitted to use AI tools like Claude and Gemini for assistance on homework assignments. This should be acknowledged explicitly with a brief statement about how AI was used.

Tentative Schedule

Week	Dates	Topics
1	Sep 10	Course Overview, History, and Review
2	Sep 15	Coordinates and Newtonian Mechanics Review
	Sep 17	State Space, Numerical Integration, Energy, and Lyapunov Stability
3	Sep 22	Work, d'Alembert's Principle, Euler-Lagrange Equation
	Sep 24	Manipulator Equation, Mass Matrix
4	Sep 29	Dynamics in Rotating Frames
	Oct 1	Linear and Angular Momentum, Rigid Bodies
5	Oct 6	Attitude Kinematics, Rotation Matrices
	Oct 8	Quaternions and Simulation
6	Oct 13	No Class — Monday Schedule
	Oct 15	Euler's Equation, Equilibria
7	Oct 20	Linearization and Stability, Spinning Spacecraft
	Oct 22	Newton-Euler Dynamics for Aircraft
8	Oct 27	Mid-Term Review
	Oct 29	**Mid-Term Exam**
9	Nov 3	Intrinsic Coordinates and Frenet-Serret (Zac Out)
	Nov 5	TBD (Zac Out)
10	Nov 10	Gravitational Potential, The Kepler Problem
	Nov 12	Orbital Elements, Perturbations, The J2 Effect
11	Nov 17	Orbital Transfers
	Nov 19	The Circular Restricted Three-Body Problem, Equilibria and Stability
12	Nov 24	3D Multi-Body Dynamics and Simulation
	Nov 26	No Class — Thanksgiving
13	Dec 1	Constraints, Lagrange Multipliers, and Constraint Stabilization
	Dec 3	Coulomb Friction and the Maximum Dissipation Principle
14	Dec 8	System Identification and Least-Squares
	Dec 10	TBD: Schedule Margin
15	Dec 14	**Final Exam** 9:00–12:00 in 35-225